Project Assignment 5

Compiler Design Lab
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Project Task G Implement Sparse Conditional Constant Propagation

For this project assignment, we introduce an additional compiler switch, `--optimize`. This switch performs the same operation as `--compile` and additionally performs optimizations. Implement the following optimization:

- Perform the SCCP [1] analysis.
- If applicable,
  - substitute instructions with constants,
  - substitute conditional branches with unconditional ones,
  - remove dead instructions and
  - remove unreachable blocks.
- If you do not want to implement SSA construction yourself, you may use the LLVM *mem2reg* pass (see `llvm::createPromoteMemoryToRegisterPass()`) **but no other LLVM passes**.
- This is the last project assignment. Information on scheduling your code review meeting with us will be made available online.
- The soft deadline for this milestone is 2023-02-10.
- Keep it simple!

[1] consider `https://dl.acm.org/citation.cfm?id=103136` for more details if you haven't already (accessible from the university network).